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Method for controlling vehicle powertrain based on obstacle detection

Abstract

A method for controlling driveline torque on an electrified powertrain based on obstacle detection is provided. The electrified powertrain includes a first eMotor, a second eMotor and an internal combustion engine (ICE). An obstacle is detected proximate to the vehicle. A proximity signal is communicated to a first torque module that determines a first torque limit. A road surface is detected. A traction limit signal is communicated to a second torque module that determines a second torque limit. A road curvature is detected. A curvature signal is communicated to a third torque module that determines a third torque limit. A safety tolerance is determined based on the first, second and third torque limit. A first torque request is communicated to the ICE. A second torque request is communicated to the first eMotor. A third torque request is communicated to the second eMotor.

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Background/Summary

FIELD

(1) The present application generally to hybrid electrified vehicles and, more particularly, to a control strategy that actively controls torque tolerances based on obstacle detection.

BACKGROUND

(2) An electrified vehicle (hybrid electric, plug-in hybrid electric, range-extended electric, battery electric, etc.) includes at least one battery system and at least one electronic drive module having an electric motor and associated electric drive gearbox assembly. Typically, the electrified vehicle would include a high voltage battery system and a low voltage (e.g., 12 volt) battery system. In such a configuration, the high voltage battery system is utilized to power at least one electric motor configured on the vehicle and to recharge the low voltage battery system via a direct current to direct current (DC-DC) convertor. Hybrid electric vehicles also include an internal combustion engine (ICE) that provides another torque input for delivering to vehicle drive wheels. A controller determines which one or combination of the electric motors and ICE is used to deliver torque to the vehicle drive wheels based on operating conditions.

(3) Many electrified vehicles include obstacle detection sensors that send signals to a vehicle controller indicative of a sensed obstacle. Existing torque control strategies rely on calibrating the system torque diagnostic to protect people during worst-case events. In examples, a worst-case

event can include detection of a pedestrian nearby in a parking lot. Such use cases make up a small portion of the actual driving experience. In this regard, such control strategies can be more sensitive than what is necessary for all driving conditions. Accordingly, while such control strategies do work well for their intended purpose, there is a desire for improvement in the relevant art.

SUMMARY

(4) According to one example aspect of the invention a method for controlling driveline torque on an electrified powertrain of a vehicle based on obstacle detection is provided. The electrified powertrain includes a first eMotor, a second eMotor and an internal combustion engine (ICE). An obstacle is detected proximate to the vehicle. A proximity signal is communicated to a first torque module that determines a first torque limit based on the detecting. A road surface is detected. A traction limit signal is communicated to a second torque module that determines a second torque limit based on the detecting. A road curvature is detected. A curvature signal is communicated to a third torque module that determines a third torque limit based on the detecting. A safety tolerance is determined based on the first, second and third torque limit. A first torque request is communicated based on the safety tolerance to an engine controller that controls torque output of the ICE. A second torque request is communicated based on the safety tolerance to a first eMotor controller that controls torque output of the first eMotor. A third torque request is communicated based on the safety tolerance to a second eMotor controller that controls torque output of the second eMotor.

(5) In some implementations, detecting an obstacle comprises detecting the obstacle with an obstacle detection sensor. Detecting a road surface comprises detecting the road surface with a traction control sensor. Detecting the road curvature comprises detecting the road curvature with a road curvature sensor.

(6) In some implementations, the first torque module determines an excess acceleration allowed before colliding with the detected obstacle and outputs the first torque limit based on the determination. The second torque module determines an excess torque allowed before wheels of the vehicle slip and outputs the second torque limit based on the determination. The third torque module determines an excess torque allowed before there is one of understeer and oversteer of the vehicle and outputs the third torque limit based on the determination.

(7) In some implementations, an engine torque request, a first eMotor torque request and a second eMotor torque request is received at a modeled system torque module. A modeled system torque is determined and communicated to a safety check module. A desired vehicle acceleration is determined. A system torque request is communicated to the safety check module based on the determining. The safety check module communicates the first, second and third torque requests based on the system torque request, the modeled system torque and the safety tolerance. The safety check module communicates the first, second and third torque requests based on the modeled torque not being greater than a sum of the system torque request and the safety tolerance.

(8) In some implementations, detecting the first, second and third torque limits comprises detecting the first, second and third torque limits at an autonomous driving controller (ADC).

Communicating the first, second and third torque requests comprises communicating the first, second and third torque limits at an electric vehicle control unit (EVCU).

(9) An electrified powertrain of a vehicle that controls driveline torque based on obstacle detection includes a first eMotor, a second eMotor and an internal combustion engine (ICE). An obstacle detection sensor detects an obstacle proximate to the vehicle and communicates a proximity signal to a first torque module that determines a first torque limit based on the detecting. A road surface sensor detects a road surface and communicates a traction limit signal to a second torque module that determines a second torque limit based on the detecting. A road curvature sensor detects a road curvature and communicates a curvature signal to a third torque module that determines a third torque limit based on the detecting. A minimum torque module determines a safety tolerance based on the first, second and third torque limit. A safety check module communicates (i) a first torque

request to an engine controller that controls torque output of the ICE, (ii) a second torque request to a first eMotor controller that controls torque output of the first eMotor; and (iii) a third torque request to a second eMotor controller that controls torque output of the second eMotor based on the safety tolerance.

(10) In additional implementations, the first torque module determines an excess acceleration allowed before colliding with the detected obstacle and outputs the first torque limit based on the determination. The second torque module determines an excess torque allowed before wheels of the vehicle slip and outputs the second torque limit based on the determination. The third torque module determines an excess torque allowed before there is one of understeer and oversteer of the vehicle and outputs the third torque limit based on the determination.

(11) In other implementations, a modeled system torque module receives an engine torque request, a first eMotor torque request and a second eMotor torque request and determines a modeled system torque based on the engine torque request, the first eMotor torque request and the second eMotor torque request. The safety check module communicates the first, second and third torque requests based on a system torque request, the modeled system torque and the safety tolerance. The safety check module communicates the first, second and third torque requests based on the modeled torque not being greater than a sum of the system torque request and the safety tolerance.

(12) Further areas of applicability of the teachings of the present application will become apparent from the detailed description, claims and the drawings provided hereinafter, wherein like reference numerals refer to like features throughout the several views of the drawings. It should be understood that the detailed description, including disclosed embodiments and drawings referenced therein, are merely exemplary in nature intended for purposes of illustration only and are not intended to limit the scope of the present disclosure, its application or uses. Thus, variations that do not depart from the gist of the present application are intended to be within the scope of the present application.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a functional block diagram of a hybrid electrified vehicle having first and second electric motors, an internal combustion engine, an autonomous driving controller (ADC) and an electric vehicle control unit (EVCU) according to various principles of the present application; and (2) FIGS. 2-4 are a logic flow diagram of a control method for controlling the hybrid electrified vehicle of FIG. 1 according to various principles of the present application.

DESCRIPTION

(3) As discussed above, many electrified vehicles include obstacle detection sensors that send signals to a vehicle controller indicative of a sensed obstacle. Existing torque control strategies rely on calibrating the system torque diagnostic to protect people during worst-case events. In examples, a worst-case event can include detection of a pedestrian nearby in a parking lot. Such use cases make up a small portion of the actual driving experience. Current torque diagnostic control strategies are tuned such that the diagnostic will react fast enough such that there is no unreasonable risk of injuring a nearby pedestrian.

(4) Typically, diagnostic tolerances are kept extremely tight during vehicle maneuvers where the vehicle may have no nearby obstacles. This can lead the system prone to false failing a safety diagnostic and triggering a reaction where there was no actual hazard. A loss of propulsion (the reaction to this diagnostic) itself can be a safety concern. In this regard, such control strategies can be more sensitive than what is necessary for all driving conditions.

(5) The instant disclosure provides an autonomous system that, instead of designing for such worst-case scenarios, is more selective in reacting based on the current actual circumstances. The present

disclosure derives tolerances based on obstacle detection from the ADC, rather than only defaulting to a worst-case scenario. The system torque diagnostic can detect issues with the sensitivity to both protect people from harm as well as remain as robust as possible. The present system uses information about the current driving scenario (obstacle detection) to derive system torque tolerances rather than basing these tolerances on engineering analyses and having them become “fixed” during the diagnostic's implementation.

(6) In examples, if the vehicle is on a highway and there are no pedestrians, the safety strategy can be adapted to be more relaxed. Further, the control strategy can determine that the tolerances on the torque commands are less important compared to a timing of how fast the safety mechanism or command is applied. In this regard, the safety strategy can be custom tuned to be based on more information about the environment which will help with its robustness.

(7) Referring now to FIG. 1, a functional block diagram of an example electrified vehicle **100** (also referred to herein as “vehicle **100**”) according to the principles of the present application is illustrated. The vehicle **100** includes an electrified powertrain **104** having an electric drive module (EDM) **106** configured to generate and transfer drive torque to a driveline **108** for vehicle propulsion. The EDM **106** includes drive motors collectively identified at **114**. The drive motors **114** can include a first electric drive units or eMotor **116A** (e.g., electric traction motor), a second eMotor **116B** and an internal combustion engine (ICE) **118**. The driveline **108** can include combinations of drive wheels and driven wheels collectively identified at reference **110**. The EDM **106** directs torque inputs from one or more of the eMotors **116A**, **116B** and ICE **118** into an electric drive gearbox assembly or transmission **120**. The transmission **120** takes the torque input and delivers it to the driveline **108**.

(8) The electric motors **116A**, **116B** are selectively connectable via a power inverter module (PIM) **124** to a high voltage battery system **112** for powering the electric motors **116A**, **116B**. The battery system **112** is selectively connectable (e.g., by the driver) to an external charging system **124** (also referred to herein as “charger **124**”) for charging of the battery system **112**. The battery system **112** includes at least one battery pack assembly **130**. A first eMotor controller **140A** can send a torque signal to an actuator **142A** that operates the first eMotor **116A**. A second eMotor controller **140B** can send a torque signal to an actuator **142B** that operates the second eMotor **116B**. A third ICE controller **140C** can send a torque signal to an actuator **142C** that operates the ICE **118**.

(9) Controllers **150** can provide various inputs to the EDM **106** related to selectively switching power inputs between the electric motors **116A**, **116B** and the ICE **118**. The controllers **150** according to the present disclosure includes an autonomous driving controller **152** and an electric vehicle control unit (EVCU) **154**. The autonomous driving controller **152** can also be referred to as an advanced driver assistance system or ADAS. The EVCU **154** can also be referred to as a vehicle propulsion controller.

(10) The controllers **150** receive various signals from sensors **160** based on vehicle operating conditions and determine a torque request to the EDM **106** based on the signals received. The sensors **160** can include any sensed vehicle information indicative of a driving condition. For example, the sensors **160** can include one or more obstacle detection sensors **160A**, a traction control sensor **160B**, and a road curvature sensor **160C**. As used herein, the traction control sensor **160B** and/or the road curvature sensor **160C** can include wheels sensors that sense wheel speeds of the wheels **110**. It will be appreciated that additional sensors may be used for providing an input to the controllers **150** for further robustness of the system within the scope of the present disclosure.

(11) With additional reference now to FIGS. 2-4, an exemplary control strategy for operating the EDM **106** according to the present disclosure is shown and generally identified at reference numeral **210**. The control strategy **210** includes first control steps **220** performed by the ADC **152** and second control steps **222** performed by the EVCU **154**.

(12) The first control steps **220** performed by the ADC **152** will now be described. At **230** control detects obstacles. In examples, the obstacles can be detected by the obstacle detection sensors

160A. The obstacle detection sensors **160A** output a proximity signal **232** to first torque module **234**. The first torque module **234** determines how much excess torque (acceleration) is allowed before colliding with the detected obstacle. The first torque module **234** outputs a first torque limit **236** to a minimum torque module **238**.

(13) At **240**, control detects road surfaces and predicts traction limits. In examples, the road surfaces and traction limits can be determined in part by the traction control sensors **160B**. The road surfaces that can be detected include, but are not limited to, snow, dirt, paved and other surfaces. The road surface detection sensors **160B** output a traction limit signal **242** to a second torque module **244**. The second torque module **244** determines how much excess torque is allowed before the vehicle wheels **110** slip or skid. The second torque module **244** outputs a second torque limit **246** to the minimum torque module **238**.

(14) At **250**, control detects road curvature. In examples, the road curvature can be determined in part by the road curvature sensor **160C**. The road curvature sensor **160C** can output a curvature signal **252** to a third torque module **254**. The third torque module **252** determines how much excess torque is allowed before there is understeer or oversteer. The third torque module **252** outputs a third torque limit **256** to the minimum torque module **238**. The minimum torque module **238** outputs a safety tolerance signal **260** based on the first, second and third torque limits **236**, **246** and **256**. The first control steps **220** performed by the ADC **152** also include a desired vehicle acceleration module **262** that generates a corresponding torque request signal **266** (FIG. 3).

(15) With additional reference now to FIG. 3, the second control steps **222** performed by the EVCU **150** will be further described. The system torque request signal **266** generated from the vehicle acceleration module **262** is received by a torque request module **310**. The torque request module **310** determines torque requests for each propulsion system actuator. In examples, the torque request module **310** determines a torque request for the first eMotor **116A**, the second eMotor **116B** and the ICE **118**. The torque request module **310** outputs torque request signals **312** based on the determination to a modeled system torque module **316**. The modeled system torque module **316** models the system torque based on the actuator commands and outputs a modeled system torque signal **318** to a safety check module **320**. The system torque request signal **266** is also received by the safety check module **320**. The safety check module **320** determines whether the method **210** will cause an unintended acceleration based on analysis of the modeled system torque signal **318**, the system torque request signal **266** and the safety tolerance signal **260**.

(16) With additional reference to FIG. 4, additional features of the second control steps **222** performed by the EVCU **154** will be described. The safety check module **320** determines whether the modeled torque is greater than the system torque request and the safety tolerance. In examples, the safety check module **320** compares the modeled system torque signal **318** and determines whether it is greater than a sum of the system torque request signal **266** and the safety tolerance signal **260**. If the safety check module **320** determines that the modeled system torque signal **318** is greater than the sum of the system torque request signal **266** and the safety tolerance signal **260**, control determines that the vehicle **104** will accelerate too much and shuts down the EDM **106** at **330**.

(17) If the safety check module **320** determines that the modeled system torque signal **318** is not greater than the sum of the system torque request signal **266** and the safety tolerance signal **260**, control sends the torque requests to each actuator **142A**, **142B**, **142C** at **340**. In examples, an ICE torque request **350** can be communicated to the ICE controller **140C**, a first eMotor request **352** can be communicated to the first eMotor controller **140A** and a second eMotor request **354** can be communicated to the second eMotor controller **140B**. The ICE controller **140C** controls torque output of the ICE **118**. The first eMotor controller **140A** controls torque output of the first eMotor **116A**. The second eMotor controller **140B** controls torque output of the second eMotor **116B**.

(18) As used herein, the term controller or module refers to an application specific integrated circuit (ASIC), an electronic circuit, a processor (shared, dedicated, or group) and memory that executes

one or more software or firmware programs, a combinational logic circuit, and/or other suitable components that provide the described functionality.

(19) It will be understood that the mixing and matching of features, elements, methodologies, systems and/or functions between various examples may be expressly contemplated herein so that one skilled in the art will appreciate from the present teachings that features, elements, systems and/or functions of one example may be incorporated into another example as appropriate, unless described otherwise above. It will also be understood that the description, including disclosed examples and drawings, is merely exemplary in nature intended for purposes of illustration only and is not intended to limit the scope of the present application, its application or uses. Thus, variations that do not depart from the gist of the present application are intended to be within the scope of the present application.

Claims

1. A method for controlling driveline torque on an electrified powertrain of a vehicle based on obstacle detection, the electrified powertrain including a first eMotor, a second eMotor and an internal combustion engine (ICE), the method comprising: detecting an obstacle proximate to the vehicle and communicating a proximity signal to a first torque module that determines a first torque limit based on the detecting; detecting a road surface and communicating a traction limit signal to a second torque module that determines a second torque limit based on the detecting; detecting a road curvature and communicating a curvature signal to a third torque module that determines a third torque limit based on the detecting; determining a safety tolerance based on the first, second and third torque limit; communicating, based on the safety tolerance, a first torque request to an engine controller that controls torque output of the ICE; communicating, based on the safety tolerance, a second torque request to a first eMotor controller that controls torque output of the first eMotor; and communicating, based on the safety tolerance, a third torque request to a second eMotor controller that controls torque output of the second eMotor.
2. The method of claim 1, wherein the detecting an obstacle comprises detecting the obstacle with an obstacle detection sensor.
3. The method of claim 1, wherein the detecting a road surface comprises detecting the road surface with a traction control sensor.
4. The method of claim 1, wherein the detecting the road curvature comprises detecting the road curvature with a road curvature sensor.
5. The method of claim 1, wherein the first torque module determines an excess acceleration allowed before colliding with the detected obstacle and outputs the first torque limit based on the determination.
6. The method of claim 1, wherein the second torque module determines an excess torque allowed before wheels of the vehicle slip and outputs the second torque limit based on the determination.
7. The method of claim 1, wherein the third torque module determines an excess torque allowed before there is one of understeer and oversteer of the vehicle and outputs the third torque limit based on the determination.
8. The method of claim 1, further comprising: receiving, at a modeled system torque module, an engine torque request, a first eMotor torque request and a second eMotor torque request; determining a modeled system torque; and communicating the modeled system torque to a safety check module.
9. The method of claim 8, further comprising: determining a desired vehicle acceleration; and communicating a system torque request to the safety check module based on the determining.
10. The method of claim 9, wherein the safety check module communicates the first, second and third torque requests based on the system torque request, the modeled system torque and the safety tolerance.

11. The method of claim 10, wherein the safety check module communicates the first, second and third torque requests based on the modeled torque not being greater than a sum of the system torque request and the safety tolerance.
 12. The method of claim 1, wherein detecting the first, second and third torque limits comprises detecting the first, second and third torque limits at an autonomous driving controller (ADC).
 13. The method of claim 1, wherein communicating the first, second and third torque requests comprises communicating the first, second and third torque limits at an electric vehicle control unit (EVCU).
 14. An electrified powertrain of a vehicle that controls driveline torque based on obstacle detection, the electrified powertrain comprising: a first eMotor, a second eMotor and an internal combustion engine (ICE); an obstacle detection sensor that detects an obstacle proximate to the vehicle and communicates a proximity signal to a first torque module that determines a first torque limit based on the detecting; a road surface sensor that detects a road surface and communicates a traction limit signal to a second torque module that determines a second torque limit based on the detecting; a road curvature sensor that detects a road curvature and communicates a curvature signal to a third torque module that determines a third torque limit based on the detecting; a minimum torque module that determines a safety tolerance based on the first, second and third torque limit; and a safety check module that communicates (i) a first torque request to an engine controller that controls torque output of the ICE, (ii) a second torque request to a first eMotor controller that controls torque output of the first eMotor; and (iii) a third torque request to a second eMotor controller that controls torque output of the second eMotor based on the safety tolerance.
 15. The electrified powertrain of claim 14, wherein the first torque module determines an excess acceleration allowed before colliding with the detected obstacle and outputs the first torque limit based on the determination.
 16. The electrified powertrain of claim 14, wherein the second torque module determines an excess torque allowed before wheels of the vehicle slip and outputs the second torque limit based on the determination.
 17. The electrified powertrain of claim 14, wherein the third torque module determines an excess torque allowed before there is one of understeer and oversteer of the vehicle and outputs the third torque limit based on the determination.
 18. The electrified powertrain of claim 17, wherein the safety check module communicates the first, second and third torque requests based on a system torque request, the modeled system torque and the safety tolerance.
 19. The electrified powertrain of claim 14, further comprising a modeled system torque module that receives an engine torque request, a first eMotor torque request and a second eMotor torque request and determines a modeled system torque based on the engine torque request, the first eMotor torque request and the second eMotor torque request.
 20. The electrified powertrain of claim 19, wherein the safety check module communicates the first, second and third torque requests based on the modeled torque not being greater than a sum of the system torque request and the safety tolerance.
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